L14: Knowledge Check

1 Multiple answer 1 point

In the node embedding problem, we compute a d-dimensional vector for each node of the network. Which of the following statements is/are true about the magnitude of d:

- [ ] d should be at least equal to the number of nodes in the network.
- [ ] The higher d is, the closer the similarity of two node embeddings to the network similarity of the corresponding nodes.
- [ ] Any value of d will produce good node embeddings but higher values of d give embeddings with higher inductive power.
- [ ] If d is not high enough, it may be that two pairs of nodes have almost the same dot product even though they are not equally similar.

2 Multiple answer 1 point

Consider the architecture of Graph Neural Networks (GNNs). Which of the following statements is/are true?

- [ ] The embedding of each node v at a layer k (k>0) is a linear function of the embeddings of node v's neighbors (including itself) at layer (k-1).
- [ ] If two nodes u and v have the same set of neighbors, then they will have the same embedding vectors at each layer.
- [ ] The dimensionality of the matrices W_k and B_k is $d_k \times d_{k-1}$, where d_k is the embedding dimensionality at layer k.
- [ ] If we do not have any input node features, we can simply set the initial embedding of each node v to zero (i.e., $h_v^0 = 0$ for all nodes v).

3 Multiple answer 1 point

Consider the cross-entropy loss function used to train GNNs for binary node classification. Which of the following statements is/are correct?

- [ ] The loss function is close to 0 whenever the model predicts correctly the label of a node.
- [ ] The loss function is equal to 1 whenever the model predicts incorrectly the label of a node.
- [ ] The loss function assumes that the activation function σ varies between 0 and 1.
- [ ] The loss function assumes that the activation function σ is equal to either 0 or 1.

4 Multiple answer 1 point

Consider the GNN we reviewed in the polypharmacy application. Which of the following statements is/are true about the "decoder" part of the network?

- [ ] It predicts whether an edge of a given type exists between any two given nodes
- [ ] It predicts whether an edge exists between two given drug nodes
- [ ] It predicts whether an edge of a given type exists between two given drug nodes
- [ ] The edge predictions use the same trainable parameters for any type of edge

5 Multiple answer 1 point

Which of the following statement(s) is/are true about the DeepWalk method?

- [ ] Ordering the nodes encountered during the random walk is important because earlier nodes receive a larger weight in the softmax ratio.
- [ ] DeepWalk embeddings can capture global information better than embeddings based on cosine similarity.
- [ ] If a node is encountered more than once in a random walk, DeepWalk filters out such duplicates to avoid bias.
- [ ] DeepWalk prefers long random walks so that even two distinct nodes co-occur in the same walk.

6 Multiple choice 1 point

The section "Deep Embeddings and Graph Neural Networks" of this Lesson described a way to compute node embeddings considering the given node features. Suppose we are also given some edge features (e.g., the strength of the connection $\mathbf{e}_{u,v}$ between nodes u and v). What can we do to consider both node and edge attributes in the computation of node embeddings?

- ◯ The approach described in this lecture already covers edge features
- ◯ We can use $\mathbf{h}_v^k = \sigma\left(\frac{1}{|N(v)|}\sum_{u \in N(v)} \phi(\mathbf{e}_{u,v})\mathbf{h}_u^{k-1} + \mathbf{B}_k\mathbf{h}_v^{k-1}\right)$ where $\phi(\mathbf{e}_{u,v})$ is a trainable neural network that maps $\mathbf{e}_{u,v}$ to $\mathcal{R}^{d_k \times d_{k-1}}$.
- ◯ We can calculate a scalar feature for each edge by adding the elements of the corresponding feature vector for that edge.
- ◯ It is not possible to consider edge features in the computation of node embeddings.